

Anton H. Westveld

Canberra, ACT, AUS & Richmond, VA, USA

anton.westveld@anu.edu.au.com | <https://awest3.github.io> | [LinkedIn](#) | [Google Scholar](#)

Affiliation

Senior Lecturer (Level C) of Statistics (Australian System)

Jan 2017 - Present

Research School of Finance, Actuarial Studies and Statistics (RSFAS)

College of Business and Economics, Australian National University

Canberra, ACT, Australia

- Convenor of the [ASA DataFest](#) at ANU [2021 - Present]
- CBE representative on the CoSM Education Committee [2020 - Present]
- RSFAS Honours Program Convenor [2015 - 2019]

Affiliate Associate Professor

Aug 2023 - Present

Department of Mathematics and Statistics

Virginia Commonwealth University

Richmond, Virginia, USA

Education

Ph.D. in Statistics, University of Washington

Seattle, WA, USA

M.A. in Applied Statistics & Applied Economics, University of Michigan

Ann Arbor, MI, USA

B.A. in Economics & Political Science, University of Michigan

Ann Arbor, MI, USA

Work Experience

Associate

Jan 2013 - Dec 2017

Statistics Laboratory @ the Bio5 Institute

and the Statistics Graduate Inter-Disciplinary Program (GIDP)

University of Arizona, Tucson, AZ, USA

Lecturer (Level B) of Statistics

Jul 2014 - Dec 2016

Research School of Finance, Actuarial Studies and Statistics

College of Business and Economics, Australian National University

Canberra, ACT, AUS

Assistant Professor of Statistics

Jan 2013 - May 2014

Faculty of ESTEM

University of Canberra

Canberra, ACT, AUS

Statistician and Regular Statistics Faculty

Sep 2011 - Nov 2012

Statistics Laboratory @ the Bio5 Institute

and the Statistics Graduate Inter-Disciplinary Program (GIDP)

University of Arizona, Tucson, AZ, USA

Assistant Professor of Statistics

Jul 2007 - Aug 2011

Department of Mathematical Sciences

University of Nevada

Las Vegas, NV, USA

Last updated: 2026-07-30

Post Doctoral Fellow Jan - Jul 2007
Center for Empirical Research in the Law
Washington University
St. Louis, MO, USA

Visiting Positions

Visiting Faculty Aug 2019 - Feb 2020
Department of Statistics and Data Sciences
University of Texas
Austin, TX, USA

Visiting Faculty Jul - Aug 2019
Center for Statistics and the Social Sciences
University of Washington
Seattle, WA, USA

Visiting Faculty Dec 2017
Department of Statistics and Data Sciences
University of Texas
Austin, TX, USA

Visiting Researcher Mar 2010
Division of Mathematics, Informatics, and Statistics
CSIRO
Canberra, ACT, AUS

Visiting Researcher Jun - Jul 2008
Division of Computational Biology
Fred Hutchinson Cancer Research Center
Seattle, WA, USA

Continuing Education

Operational AI: Balancing National Security & Ethical Governance Feb - Mar 2025
Whole of Government Center of Excellence and the Law School
William & Mary
Williamsburg, VA, USA

Modern Leadership: Reflections and Tools for the Values-Based Leader May 2021
William & Mary Washington Center
Washington, DC, USA

**Preparing Statisticians for Leadership:
How to See the Big Picture and Have More Influence** Jul 2018
JSM
Vancouver, BC, CAN

Scientific Publications

Marin[#], Santiago, Bronwyn Loong, and Anton H. Westveld. “Adaptive Shrinkage with a Nonparametric Bayesian Lasso”. Preprint: [arXiv:2411.08262](https://arxiv.org/abs/2411.08262), *Journal of Computational and Graphical Statistics* (2026). <https://doi.org/10.1080/10618600.2025.2572327>.

Marin[#], Santiago, Bronwyn Loong, and Anton H. Westveld. “BOB: Bayesian Optimized Bootstrap with Applications to Gaussian Mixture Models”. Preprint: [arXiv:2311.03644](https://arxiv.org/abs/2311.03644), *Statistics and Computing* (2026). <https://doi.org/10.1007/s11222-025-10763-y>.

- Wang[#], Jiali, Anton H. **Westveld**, Alan H. Welsh, Melissa Parker, and Bronwyn Loong. “Analysis of Survey on Menstrual Disorder Among Teenagers Using Gaussian Copula Model with Graphical Lasso Prior”. *PLOS ONE* 3 (2021): 1–22. <https://doi.org/10.1371/journal.pone.0248340>.
- Lundy, Michelle B., Cynthia A. Standley, and Anton H. **Westveld**. “Curricular Reform in Two Medical School Tracks and the Impact on USMLE Scores”. *Medical Science Educator* 27, no. 2 (2017): 201–207. <https://doi.org/10.1007/s40670-016-0368-y>.
- Chiu, Grace S., and Anton H. **Westveld**. “A Statistical Social Network Model for Consumption Data in Trophic Food Webs”. Preprint: <https://arxiv.org/abs/1006.4432>, *Statistical Methodology (Special issue on Modern Statistical Methods in Ecology)* 17 (2014): 139–160. <https://doi.org/10.1016/j.stamet.2013.09.001>.
- Hurwitz, Bonnie L., Anton H. **Westveld**, Jennifer R. Brum, and Mathew B. Sullivan. “Modeling Ecological Drivers in Marine Viral Communities Using Comparative Metagenomics and Network Analyses”. *Proceedings of the National Academy of Sciences of the USA* 111, no. 29 (2014): 10714–10719. <https://doi.org/10.1073/pnas.1319778111>.
- Taylor, Amber, et al. “The Use of Metformin is Associated with Decreased Lumbar Radiculopathy Pain”. *Journal of Pain Research* 2013, no. 6 (2013): 755–763. <https://doi.org/10.2147/JPR.S52205>.
- Westveld**, Anton H., and Peter D. Hoff. “Modeling of the Learning Process in Centipede Games”. Preprint: [arXiv:1003.2253](https://arxiv.org/abs/1003.2253), *Stat* 2, no. 1 (2013): 242–254. <https://doi.org/10.1002/sta4.32>.
- Chiu, Grace S., and Anton H. **Westveld**. “A Unifying Approach for Food Webs, Phylogeny, Social Networks, and Statistics”. *Proceedings of the National Academy of Sciences of the USA* 108, no. 38 (2011): 15881–15886. <https://doi.org/10.1073/pnas.1015359108>.
- Chiu., Grace S., Peter Guttorp, Anton H. **Westveld**, Shahedul A. Khan, and Jun Liang. “Latent health factor index: a statistical modeling approach for ecological health assessment”. *Environmetrics* 22, no. 3 (2011): 243–255. <https://doi.org/https://doi.org/10.1002/env.1055>.
- Westveld**, Anton H., and Peter D. Hoff. “A Mixed Effects Model for Longitudinal Relational and Network Data, with Applications to International Trade and Conflict”. Preprint: [arXiv:1009.1436](https://arxiv.org/abs/1009.1436), *Annals of Applied Statistics* 5, no. 2A (2011): 843–872. <https://doi.org/10.1214/10-A0AS403>.
- Gneiting, Tillman, Adrian E. Raftery, Anton H. **Westveld**, and Tom Goldman. “Calibrated Probabilistic Forecasting Using Ensemble Model Output Statistics and Minimum CRPS Estimation”. *Monthly Weather Review* 133, no. 3 (2005): 1098–1118. <https://doi.org/10.1175/MWR2904.1>.

[#] denotes a student that I supervise(d) in some manner.

Working Papers

- Alrawkan[#], Riyadh, Edward Boone, Ryad Ghanam, and Anton H. **Westveld**. “Linear Models, Variable Selection, Artificial Intelligence”. *arXiv Statistics Methodology* (2026). **Under Review**, <https://doi.org/10.48550/arXiv.2604.27191>.
- Marin[#], Santiago, Bronwyn Loong, and Anton H. **Westveld**. “Bayesian nonparametric modeling of dynamic pollution clusters through an autoregressive logistic-beta Stirling-gamma process”. *arXiv Statistics Methodology* (2026). **Under Revision**, <https://doi.org/10.48550/arXiv.2601.04625>.

- Prevett[#], Mitchell L., Francis K. C. Hui, Zhi Yang Tho, A. H. Welsh, and Anton H. **Westveld**. “Gradient Boosted Mixed Models: Flexible Estimation of Mean and Variance Components for Clustered Data”. *arXiv Statistics Methodology* (2026). **Under Review**, <https://doi.org/10.48550/arXiv.2511.00217>.
- Chiu, Grace S., et al. “Model-based Calibration of Gear-Specific Fish Abundance Survey Data as a Change-of-Support Problem”. *arXiv Statistics Methodology* (2025). **Under Revision**, <https://doi.org/10.48550/arXiv.2505.05767>.
- Wolff[#], Malcolm, Adrian Dobra, Anton H. **Westveld**, and Grace S. Chiu. “Modeling Human Spatial Mobility Patterns with the Lévy Flight Cluster Model”. *arXiv Statistics Methodology* (2025). **Under Revision**, <https://doi.org/10.48550/arXiv.2509.00298>.
- Kuh[#], Fui S., Grace S. Chiu, and Anton H. **Westveld**. “Latent Causal Socioeconomic Health Index”. *arXiv Statistics Methodology* (2023). **Under Revision**, <https://arxiv.org/abs/2009.12217>.
- Kuh[#], Fui S., Grace S. Chiu, and Anton H. **Westveld**. “Modeling National Latent Socioeconomic Health and Examination of Policy Effects via Causal Inference”. *arXiv Statistics Applications* (2019). <https://arxiv.org/abs/arXiv:1911.00512>.
- Wang[#], Jialia, Anton H. **Westveld**, Bronwyn Loong, and Alan H. Welsh. “An Imputation model by Dirichlet Process Mixture of Elliptical Copulas for Data of Mixed Type”. *arXiv Statistics Methodology* (2019). <https://arxiv.org/abs/1910.05473>.
- Wang[#], Jiali, Anton H. **Westveld**, Bronwyn Loong, and Alan H. Welsh. “A Copula-based Imputation Model for Missing Data of Mixed Type in Multilevel Data Sets”. *arXiv Statistics Methodology* (2017). <https://arxiv.org/abs/1702.08148>.

Conference Papers

- Chien[#], Chien-Hung, Armin Haller, and Anton H. **Westveld**. “Firm Business Networks”. *International Workshops on Contextualized Knowledge Graphs, and Semantic Statistics co-located with 17th International Semantic Web Conference* (2018). <http://ceur-ws.org/Vol-2317/>.
- Chiu, Grace S., and Anton H. **Westveld**. “Physical Statistics or Statistical Physics? A Brief Overview of Bayesian Melding”. ISBN: 978-0-9872143-3-1, *J. Piantadosi and R.S. Anderssen and J. Boland (eds) MODSIM2013, 20th International Congress on Modelling and Simulation. Modelling and Simulation Society of Australia and New Zealand* (2013): 2506–2512. <https://www.mssanz.org.au/modsim2013/A7/chiu.pdf>.
- Jensen, Nathan M., Andrew D. Martin, and Anton H. **Westveld**. “Foreign Direct Investment: A Longitudinal Social Network Approach”. *Political Networks Conference* (2011). http://opensiuc.lib.siu.edu/pnconfs_2010/51/.
- Myroshnychenko[#], Maxym V., and Anton H. **Westveld**. “Analysis of Morris Water Maze Data with Bayesian Statistical Methods”. [Poster associated with the paper](#) [joint with Dr. Jefferson Kinney]. *UNLV McNair Scholars Institute Annual Report 2011-2012, University of Nevada Las Vegas* (2011): 65–71. https://digitalscholarship.unlv.edu/cgi/viewcontent.cgi?article=1002&context=mcnair_journal.
- Quinn, Kevin M., and Anton H. **Westveld**. “Bayesian Inference for Semi-parametric Quantal Response Equilibrium Models”. *JSM Proceedings, Section on Bayesian Statistical Science* (2009). <https://github.com/awest3/awest3.github.io/blob/main/docs/extraFiles/SpqreJsmProc09.pdf>.

Letnes, Bjorn, and Anton H. **Westveld**. “Transnational Corporations, Economic Development and Human Rights: On the Importance of Assets and Institutions”. *APSA Proceedings* (2005). <https://convention2.allacademic.com/one/apsa/apsa05/>.

Westveld, Anton H., and Peter D. Hoff. “Statistical Methodology for Longitudinal Social Network Data”. *APSA Proceedings* (2005). <https://convention2.allacademic.com/one/apsa/apsa05/>.

Technical & Client Reports

Chiu, Grace S., Anton H. **Westveld**, Mark A. Albins, and Allison L. White. “Overview of Bayesian Approach”. *Greater Amberjack Count Final Technical Report* (2026). <https://masgc.org/greater-amberjack>.

Chien[#], Chien-Hung, Alan H. Welsh, and Anton H. **Westveld**. “Australian Business Networks”. *ABS Research Paper* (2019). <https://www.abs.gov.au/ausstats/abs@.nsf/mf/1351.0.55.063>.

Chiu, Grace S., Shawn Treier, Anton H. **Westveld**, and Nicholas Erskine. “Progress Report on ‘Evidence-based inference for network relations using latent space network modelling: Case studies from the Australian political landscape’”. *Report prepared for ANU College of Arts and Social Sciences* (2019).

Macatangay, Kelly, Anton H. **Westveld**, and B. Kunkel. “Intervening for Success”. *Report prepared for Clark County District Attorney, Family Support Division and the U.S. Department of Health and Human Services* (2012).

Competitive Grants & Other Funding

Consulting at the [Virginia Institute of Marine Science](#) (William & Mary). 2022
 • Statistical Consultant (Bayesian methodology) - [U.S. Greater Amberjack Research Program](#).

Instructor at the [Australian Institute of Health and Welfare](#). 2019
 • One day statistics and data science with R workshop.

International Collaborator of the [SFI Centre for Research Training Foundations of Data Science](#). 2019

Co-PI of ANU RSFAS & RSSS cross-college grant. 2018
 • Evidence-based inference for network relations using latent space network modelling: Case studies from the Australian political landscape.
 • In collaboration with the other PIs: Dr. Shawn Treier and Dr. Grace Chiu.

Co-PI for sustainable farming practices from the Ian Potter Foundation. 2018
 • In collaboration with Professor David Lindenmayer (PI) et al., including Associate Professor Emma Schultz, Dr. Grace Chiu (CBE Co-PIs).

Co-PI for sustainable farming practices pilot project - 1 year grant. 2016
 • In collaboration with Professor David Lindenmayer (PI) et al., including Associate Professor Emma Schultz, Dr. Grace Chiu (CBE Co-PIs).

Co-PI for IBISWorld philanthropic grant. 2016-2018
 • IBISWorld, Melbourne, Australia - 3 year grant; Part of this grant was used to create the IBISWorld Ph.D. Scholarship awarded to Ms. F. Swen Kuh, who is jointly advised by myself and Dr. Grace Chiu (PI).

Sub-grant PI for Clark County District Attorney’s Office Family Support Division, Nevada, USA. 2009-2012

- Child Support Policies via Randomized Experiments (Evidenced Based Policy) funded from the US Department of Health & Human Services.

Sub-grant PI of NSERC Discovery Grant. 2009

- Statistical modelling of health indices for aquatic ecosystems.

PI of the New Researchers' US Federal Stimulus Award. 2008

PI of the Nevada System of Higher Education, NSF EPSCoR RING TRUE. 2007 - 2008

Courses Taught, Convened⁺, & Developed^{*}

Statistical Learning⁺⁺ 2015 - 2017, 2021 - Present

ANU STAT 3040/4040/7040 [undergraduate, honours, graduate - some years graduate and undergraduate were not jointly taught]

Statistical Inference⁺ 2014 - 2020, 2024 - Present

ANU STAT3013/4027/8027 [undergraduate, honours, graduate]

Statistics for Data Analysis A & B⁺⁺ 2021

ANU Micro-credential Course [professional]

Regression Modelling + Introduction to GLMs⁺ 2020

ANU STAT 2008/2014/6014 [undergraduate and graduate]

Regression Modelling⁺ 2018 [944 students enrolled]

ANU STAT 2008/6038 [undergraduate and graduate]

Managing the Research Process⁺⁺ 2016 - 2018

ANU STAT 4010 [honours]

Introduction to Mathematical Statistics⁺ 2014

ANU STAT2001/6039 [undergraduate and graduate]

Econometrics⁺ 2014

UC 6541/6551 [undergraduate and graduate]

Problem Evaluation and Resolution⁺ 2013 - 2014

UC 8730 [undergraduate]

Introduction to Business Statistics⁺ 2013

UC 5123/6275 [undergraduate and graduate]

Aided in developing a course on **network data**^{*} 2012

for the School of Information: Science, Technology, and Arts (SISTA)

UA [undergraduate]

Applied Statistics for the Biological Sciences⁺ 2008 - 2010

UNLV STAT 391 [undergraduate]

Experimental Design and Analysis⁺ 2008 - 2010

UNLV STAT 713 [graduate]

Bayesian Data Analysis⁺⁺ 2009

UNLV STAT 753 [graduate]

Independent study for **Bayesian Data Analysis**⁺⁺ 2008 - 2011

UNLV STAT 753 [graduate]

Regression Analysis II⁺⁺ (GLMs, LMMs, GLMMs) 2010 - 2011

UNLV STAT 762 [graduate]

Last updated: 2026-07-30

Statistical Decision Theory⁺ UNLV Stat 765 [graduate]	2008
Multivariate Analysis⁺ UNLV Stat 764 [graduate]	2007

Note: Australian National University (ANU), University of Arizona (UA), University of Canberra (UC), University of Nevada, Las Vegas (UNLV)

Research Advising and Mentoring

Mitchell Prevett , Ph.D. student in Statistics, RSFAS, ANU. Project: Gradient Boosting for Mixed Models	2025 - Present Co-Adv
Riyadh Alrawkan , Ph.D. student in Operations Research, Dept. of Mathematics and Statistics, VCU. Project: Artificial Intelligence and Statistics	2026 - Present Co-Adv
Santiago Marin Ardila , Ph.D. student in Statistics, RSFAS, ANU. - Project: Bayesian Posterior Approximation and Non-Parametrics - Best Poster Award, 18th World Meeting of the International Society for Bayesian Analysis, June 2026 - Best Poster Award, Bayes on the Beach, February 2024	2023 - Present Co-Pri Adv
Chris Malone , Master's in Statistics, RSAF, ANU. - Project: Bayesian Inference of Eradication of Incipient Tephritid Fruit Fly - LinkedIn: https://www.linkedin.com/in/cm-ds/	2021 - 2022 Co-Pri Adv
Malcolm Wolff , Ph.D. student in Statistics, Univ. of Washington, Seattle. - Project: Statistical Modelling of Human Movement, Social Activity Spaces, and Networks - LinkedIn: https://www.linkedin.com/in/malcolm-wolff-82b1586a/	2021 - Present Co-Adv
Fui Swen Kuh , IBISWorld Ph.D. Scholar in Statistics, RSFAS, ANU. - Project: A Holistic Bayesian Framework for Modelling Latent Socio-Economic Health - Winner of the Inaugural Poster competition of the ASA Social Statistics Section - JSM 2018 - LinkedIn: https://www.linkedin.com/in/swenkuh/	2016 - 2022 Co-Pri Adv
Joseph Chien , Ph.D in Statistics, Mathematical Sciences Institute, ANU. - Project: Using Administrative Data to Gain Insights into Microdrivers of Productivity - LinkedIn: https://www.linkedin.com/in/joseph-chien-hung-chien-6a5184a9/	2016 - 2021 Assoc Adv
Jiali Wang , Ph.D in Statistics, RSFAS, ANU. - Project: Recent Developments of Copula-Based Models to Handle Missing Data of Mixed-Type in Multivariate Analysis - Recipient of the 2018 Dennis Trewin Prize presented by the Statistical Society of Australia - Canberra Branch. - LinkedIn: https://www.linkedin.com/in/jiali-wang-89244513b/	2015 - 2018 Co-Pri Adv

- Mark Dawkins**, Honours in Statistics (H1), RSFAS, ANU. 2017
Co-Pri Adv
- Project: [Melded Bayesian Inference for Stochastic Theoretical Models with Applications in Agent Based Modelling](#)
 - LinkedIn: <https://www.linkedin.com/in/mark-dawkins-479762122/>
- Jaron Lee**, Honours in Statistics (H1), RSFAS, ANU. 2016
Pri Adv
- Project: [A Variational Bayes Approach to Clustered Latent Preference Models for Directed Network Data](#)
 - Recipient of an [ANU University Medal](#)
 - Received Ph.D. in Computer Science from Johns Hopkins University
 - LinkedIn: <https://www.linkedin.com/in/jaron-jr-lee/>
- Fui Swen Kuh**, ANU Summer Research Scholar from the Department of Statistics, University of Auckland. 2015
Pri Co-Adv
- Project: [Social Network Analysis on Food Web and Dispute Data](#)
 - LinkedIn: <https://www.linkedin.com/in/swenkuh/>
- Seoyoon Cho**, CSIRO Industrial Trainee from the Department of Statistics, Korea University. 2014
Co-Adv
- Project: Modeling Network Data Using PyStan
 - Received Ph.D. in Biostatistics from University of North Carolina
- Hyangki Lee**, CSIRO Industrial Trainee from the Department of Statistics, Korea University. 2014
Co-Adv
- Project: Modeling Network Data Using PyStan
- Yaroslav Mukhin**, M.S. in Statistics, University of Nevada, Las Vegas. 2010 - 2012
Mentor
- Project: Inference for the Solow Growth Model: A Bayesian Melding Approach
 - Received Ph.D. in Economics from Massachusetts Institute of Technology
 - LinkedIn: <https://www.linkedin.com/in/yaroslav-mukhin-6b7890229/>
- Maxym Myroshnychenko**, NSF Biomathematics Scholar and UNLV McNair Scholar, University of Nevada, Las Vegas. 2010 - 2012
Pri Adv
- Project: [Analysis of Morris Water Maze Data with Bayesian Statistical Methods](#)
 - Received Ph.D. in Neuroscience from Indiana University Bloomington
 - LinkedIn: <https://www.linkedin.com/in/maxym-myroshnychenko-b79b2b51/>

[Internship \(Consulting\) Advising & Mentoring](#)

- Rory Williams**, ANU College of Business and Economics Internship Program. 2021
Pri Adv
- Joint internship through Australian Department of Social Services
- James Kondilios**, ANU College of Science Internship Program. 2019
Pri Adv
- Joint internship through Airservices Australia
- Jizhi Li**, ANU College of Science Internship Program. 2017
Pri Adv
- Joint internship through the National Disability Insurance Agency, Australia
- Yiran Yao**, ANU College of Science Internship Program. 2017
Pri Adv
- Joint internship through the National Disability Insurance Agency, Australia

Selected Professional Activities & Awards

- Associate Editor** for the quarterly bulletin of the International Society for Bayesian Analysis. 2016 - Present
- Associate Editor** for the Journal of the American Statistical Association. 2019 - 2025
- Instructor** for the statistics/data science workshop for the National Youth Science Foundation of Australia. 2019
- Best Poster Award.** Young Statisticians' Meeting (satellite meeting of the ISI), Dublin, IRL. 2011
- Title: Dirichlet Processes for Quantal Response Equilibrium Models - Examination of Pegged Exchange Rate Policies
- Best Poster Award.** Political Networks Conference, Cambridge, MA, USA. 2011
- Title: A Bayesian Mixed Effects Model for Longitudinal Social Network Data
- Student Paper Award.** Joint Statistical Meeting, Salt Lake City, UT, USA. 2007
- ASA Section on Bayesian Statistical Science
- Title: A Bayesian Mixed Effects Model for Longitudinal Social Network Data

Recent Seminars

- Invited.** Professor Adrian Raftery's [ABCDE Retirement Conference](#), University of Washington, 2024
Seattle, WA, USA.
- Modelling National Latent Socioeconomic Health and Examination of Policy Effects
- Invited.** Virginia Chapter of the ASA Conference, Charlottesville, VA, USA. 2024
- Bayesian Optimized Bootstrap (BOB) for Uncertainty Quantification in Gaussian Mixture Models
- Invited.** The International Environmetrics Society, Adelaide, AUS. 2024
- A Bayesian Melding Approach for Stochastic Agent-Based Models
- Invited.** Virginia State University, Petersburg, VA, USA. 2023
- A Bayesian Melding Approach for Stochastic Agent-Based Models
- Invited.** Virginia Commonwealth University, Richmond, VA, USA. 2023
- A Bayesian Melding Approach for Stochastic Agent-Based Models

Professional Societies

- American Statistical Association** 2002 - Present
- American Political Science Association** 2002 - Present
- Statistical Society of Australia** 2010 - Present
- International Society for Bayesian Analysis** 2017 - Present